on report

Autonomus Phisical AI & ML
for Critical Oil Pipelines
Infrastructures



Pipeline Leak Risk Management



ESG

Shifting from Reactive Monitoring
to Predictive, Self-Healing Oil
Pipeline Integrity



The Problems

Limits Legacy Monitoring

1

Reactive

Detects failures after they happen.

2

Slow

Human-in-the-loop response takes 15-45 min.

3

Blind

High false-positive rates; misses micro-fractutes.

$1.5M+
$1.2B+

4

Costly

Average hazardous liquid incident costs $1.5M; catastrophic spills exeed $1.2B.

Currently, legasy systems rely on mass-balance thresholds and human operators. If a pipe ruptures, it takes minutes for the system to alert a human, and more minutes for that human to decide to close a valve. In that 30-minutes window, a micro-fractures becomes a catastrophic environmental disaster. We are leaving millions of dollars and our environmental reputation on the table.

Proposed Solution

Our solution is a Phisical AI Platform. It doesn't just watch the pipe. It perceives the physical world throw 11 different sensor modalities, reasons about the phisics of the fluid and autonomously takes physical action to mitigate threats in under 5 sec, without waiting for human intervention.



The Paradigm Shift: Phisical AI. From Monitoring to Perceiving, Reasonong and Acting.

1

11 Modal Sensor Fusion

Combining Acoustic, Pressure, Coriolis, Fiber DTS, Temperatures, Soil Moisture, Capacitance, Vibration, Corrosion, Flow and Physics Residual data.

2

Phisics-Informed Neural Networks: PhisicsNeMo

Using fluid dynamics equations to detect anomalies that pure data-driven AI might miss.

3

Multi-Agent Reasoning: Swarm of AI agents

Sentinel for perseptions, Commander for reasoning, Operator for execution. Edge-to-edge communication via ZeroMQ and Graph Neural Network to predict cascading failures.

High-Level Architecture

C4Model

The architecture is distributed. The “brain” lives on the Edge, on Nvidia Jetson Orin devices at every pipeline node. This ensures that even if the cloud connections is lost, the pipeline protects itself. The cloud handles heavy lifting like global model training, risk assessment and generating regulatory reports.

Edge (Jetson Orin)

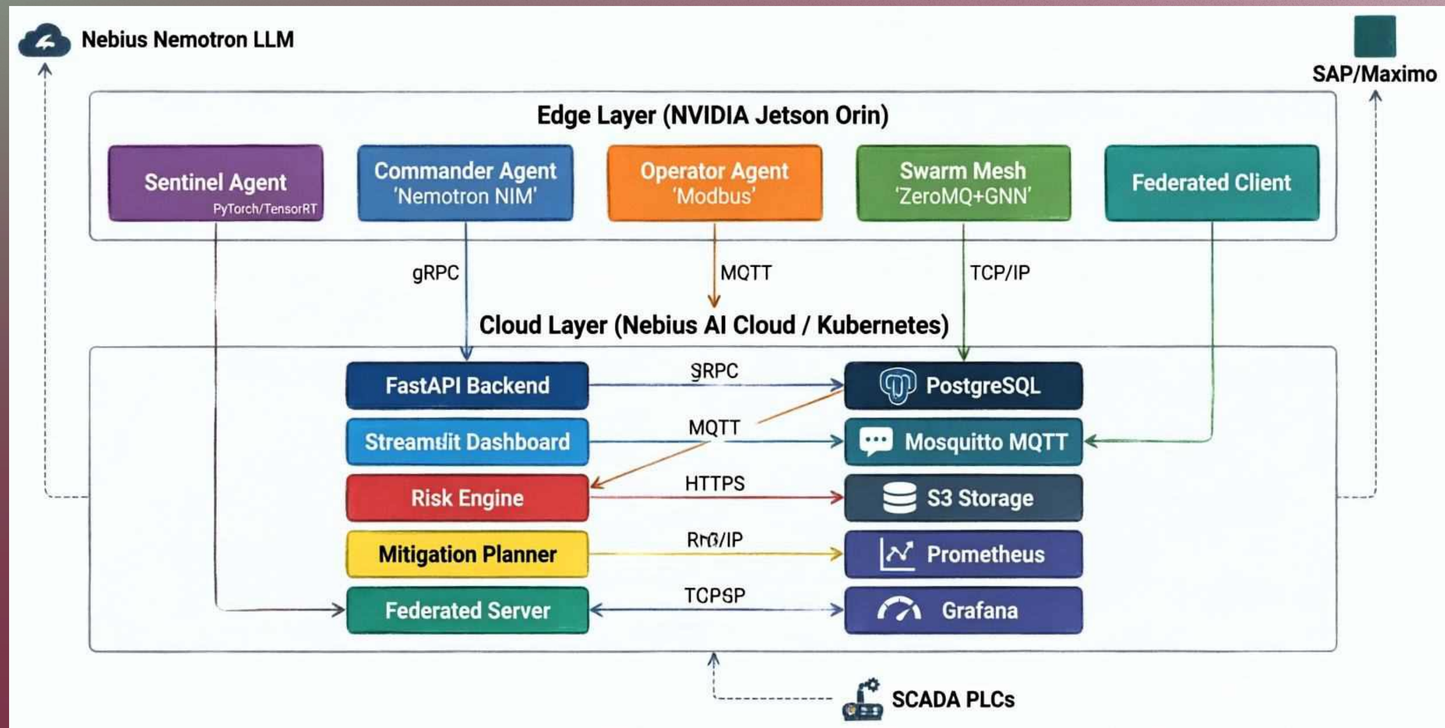
Real-time perceptions, autonomus control, 72-hours offline survival.

Cloud (Nebius AI)

Federated learning, enterprise integrations (SAP/PHMSA), digital twin.

Ingestion Pipeline:

- Phisycal Sensors - MQTT Publisher;
- MQTT Broker - Edge Agregator;
- Data Normaliser - API 1163 standards;
- PhysicsNeMo Surrogate - Computes fluid dynamics residual;
- Sequence Buffer - Sliding window;
- Sentinel Agent-PyTorch Transformer Autoencoder inference;
- Commander Agent - Nemotron NIM Reasoning + Safety Interlock validation;
- Operator Agent - Modbus TCP/OPC-UA Scada command translation.



11-Modal Fusion

Physics-Informed AI



10 Physical Sensors

Acoustic, Pressure, Coriolis, Fiber DTS, Temp, Capacitance, Soil, Vibration, Corrosion, Flow.

1 Virtual Sensor

PhysicsNeMo Residual (Navier-Stokes equations).

The Platform fuse data from 10 physical sensors. But the true breakthrough is the 11th modality: the Physics Residual. We embed the actual fluid dynamics equation - the Navier-Stokes equations - directly into the AI. The AI doesn't just look for data spikes; it looks for violations of the law of physics. This reduce false alarms by 40 times compared to legacy systems.

98.10% Precision 0.08 Falce Positives/ Day

Sentinel

Perception

Detects anomaly.

Commander

Reasoning

Proposed action via LLM.

Safety Interlock

Predictable Check

Deterministic Validation.

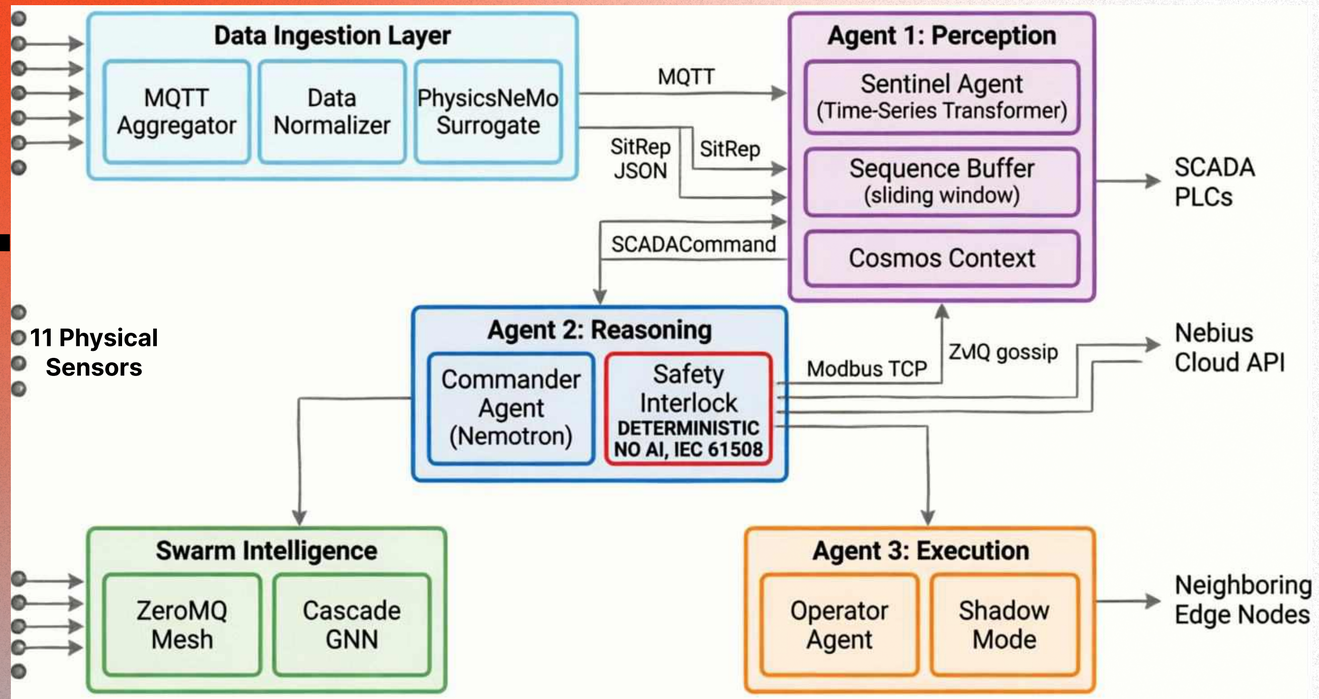
Operator

SCADA

Executes Modbus.

This is a Multi-Agent system where the AI proposes an action, but a hard coded, deterministic Safety Interlock validates it. If the AI suggests closing a valve when the pressure is too high, which could cause a water-hammer rupture - the Interlock overrides the AI and triggers a mechanical emergency shutdown. It is IEC 61508 SIL-2 compliant.

Multi-Agent Autonomy



Swarm Intelligence & Federated Learning

Data Science

Swarm Mesh

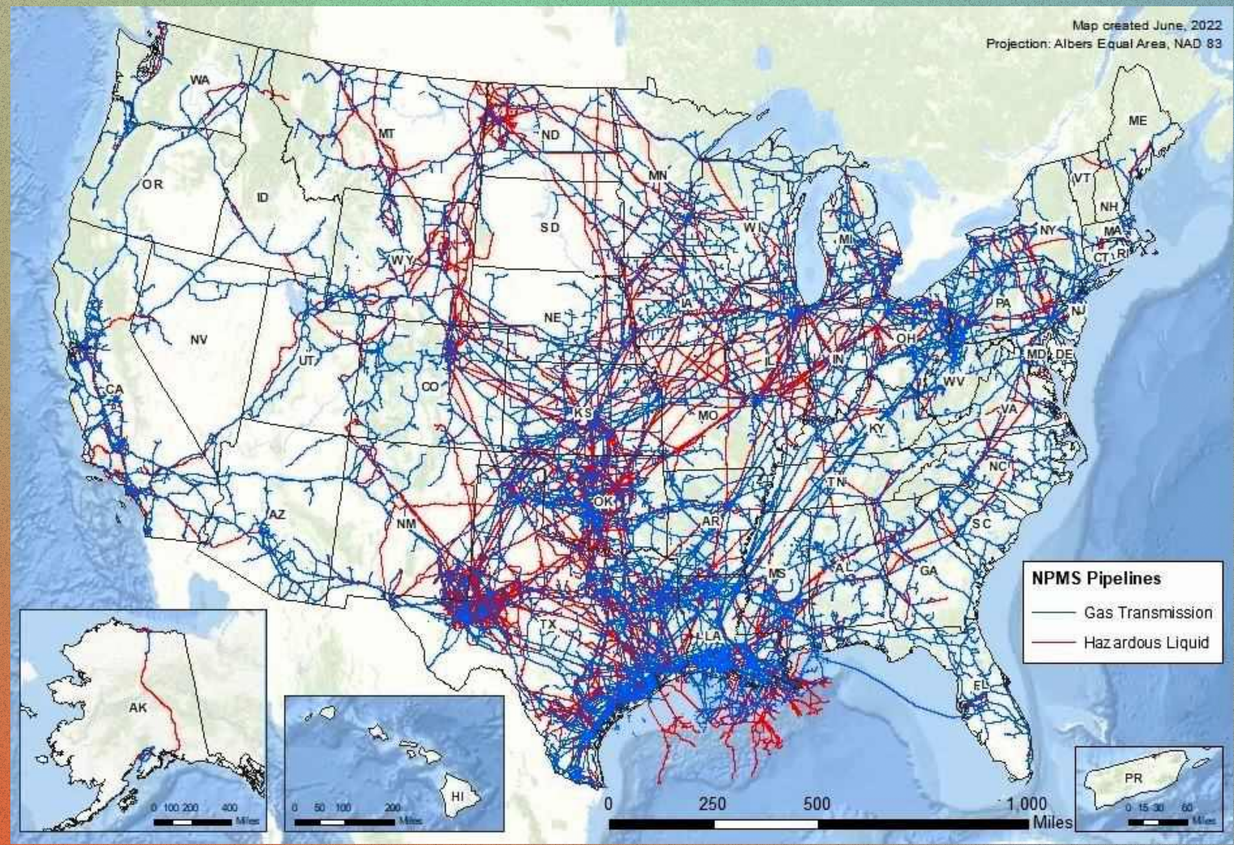
Edge-to-edge cascade prediction (Graph Neural Networks).

Federated Learning

Global AI training without raw data leaving the edge.

Zero-Trust Security

Mutual TLS encryption everywhere.



Pipelines are networks. Our Swarm Intelligence uses Graph Neural Networks to predict this cascade and pre-emptively isolate Emergency Node before the wave even arrives. Furthermore, we use Federated Learning. The raw, proprietary sensor data never leaves the edge device. We only send encrypted mathematical weight updates to the cloud, solving the industry's biggest data privacy concern.

Dynamic Risk Engine & Enterprise Integration

Streamlit

API RP 581 Risk Based Inspection

Dynamic Risk Index (PoF × CoF).

Automated Enterprise Action

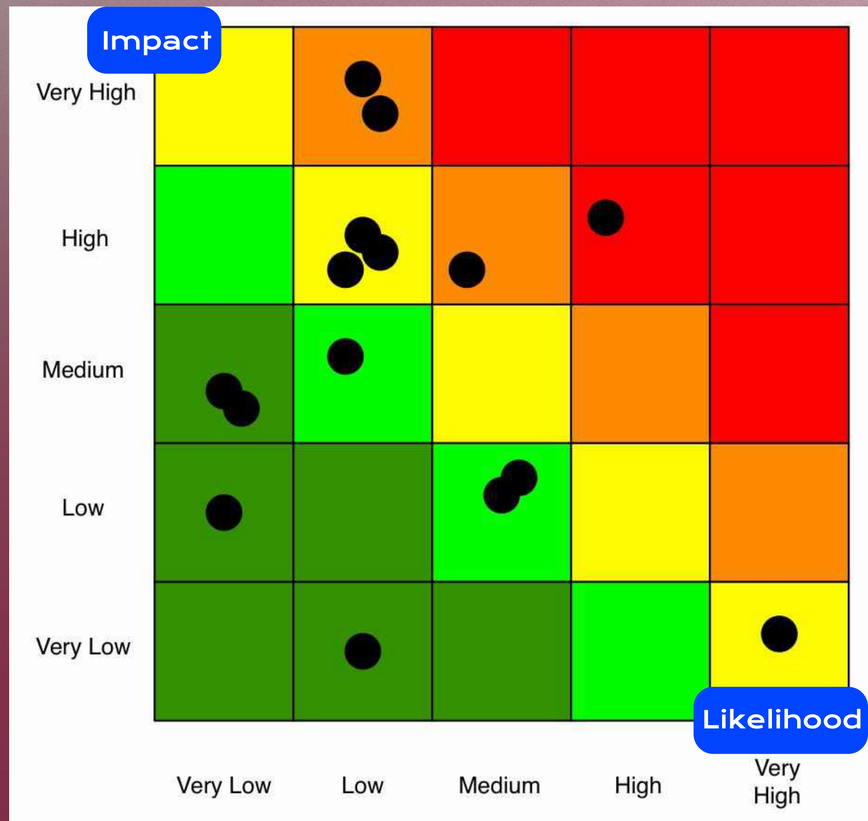
SAP/Maximo Work Orders generated instantly.

Regulatory Compliance

Automated PHMSA incident reports.



While the physical valve is closing, the cloud is working. The Risk Engine instantly recalculates the financial and environmental risk. Mitigation Planner automatically books a maintenance crew in SAP, and our LLM drafts the federal PHMSA compliance report. We turn a chaotic emergency into a structured, auditable workflow.



Business Impact & ROI

MARKET



95%

Reduction in spill volume

By detecting leaks at 0.5% deviation and acting in seconds, Operators reduce spill oil volume.

40x

Fewer false alarms

Our 11-modal fusion reduces false positives by an estimated 80%, saving in OpEx annually.

20%

Reduction in inspection OpEx

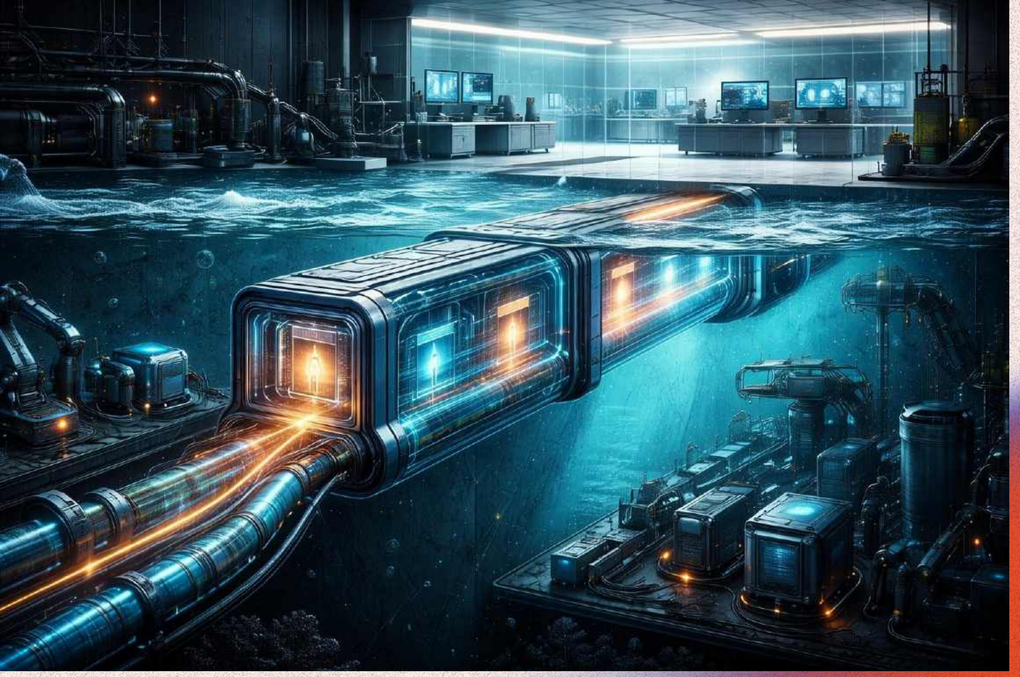
Dynamic Risk Engine tells Operators exactly which 5% of the pipe needs in-line inspection OpEx.

10%

Insurance & Regulatory Benefits

Insurance Premiums, PHMSA Compliance, improves the company's ESG and Carbon reitings.

Preventing just one major spill pays for the platform's deployment across the entire network for a decade.



Security, Compliance& Deployment

AES 256 GCM

Cybersecurity

Mutual TLS, Zero-Trust, Edge-to-Cloud Encryption.

Compliance

IEC 61508 (Safety), PHMSA (Regulatory), API 581 (Risk).

Deployment

Containerized (Docker/K8s), Terraform IaC, 1-click Scaling.



This Platform is built for the enterprise. It's secured with mutual TLS, compliant with PHMSA and IEC 61508 safety standards, and deployed via Kubernetes and Terraform.

on report

Thank You



Summary: Autonomous, Safe, Predictive, Compliant. We have built a system that perceives, reasons, and acts autonomously, bounded by deterministic safety rules, and integrated directly into our enterprise business systems. Platform protects our people, our environment, and our bottom line. This isn't just a software upgrade - it's a fundamental shift in how we protect critical infrastructure.



ESG